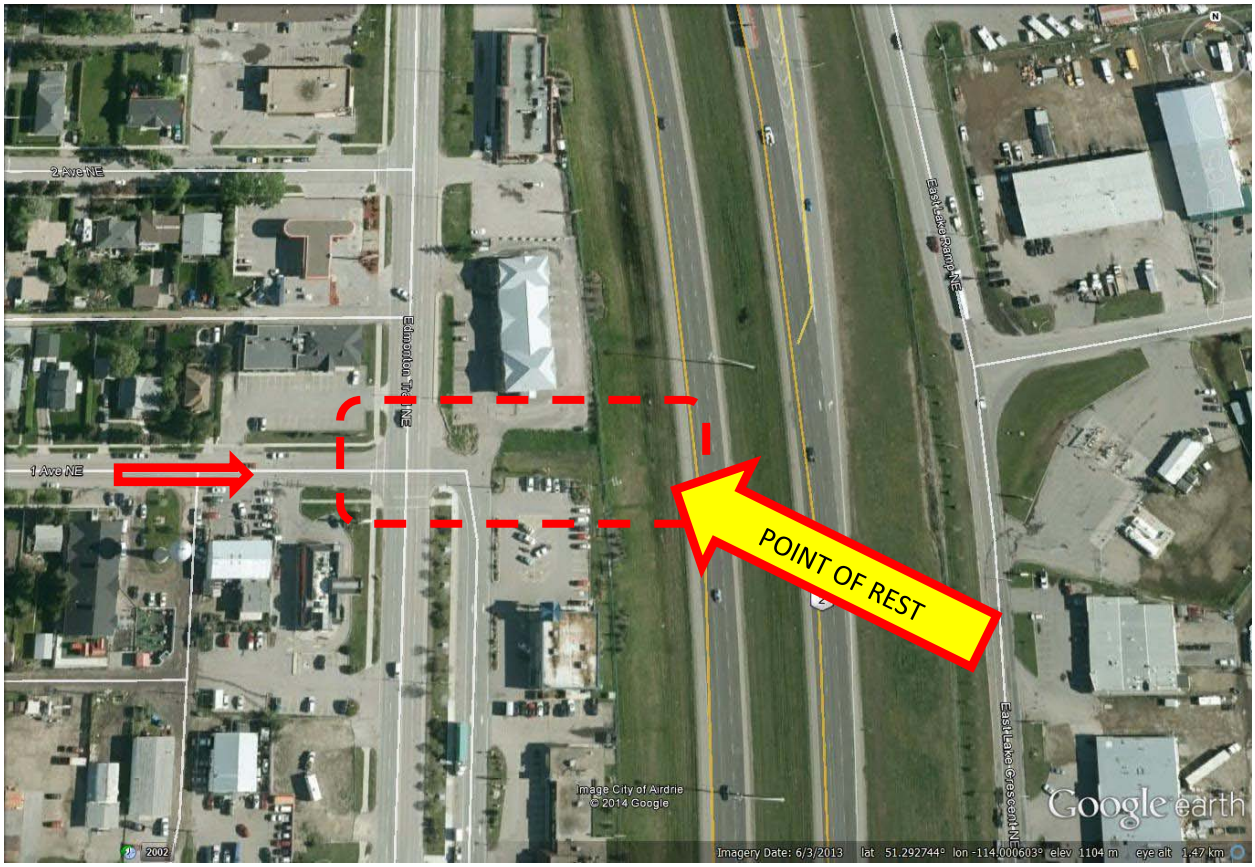
Case Study: 2014 Tacoma “13 EDR”: Fleeing Police Ends Badly

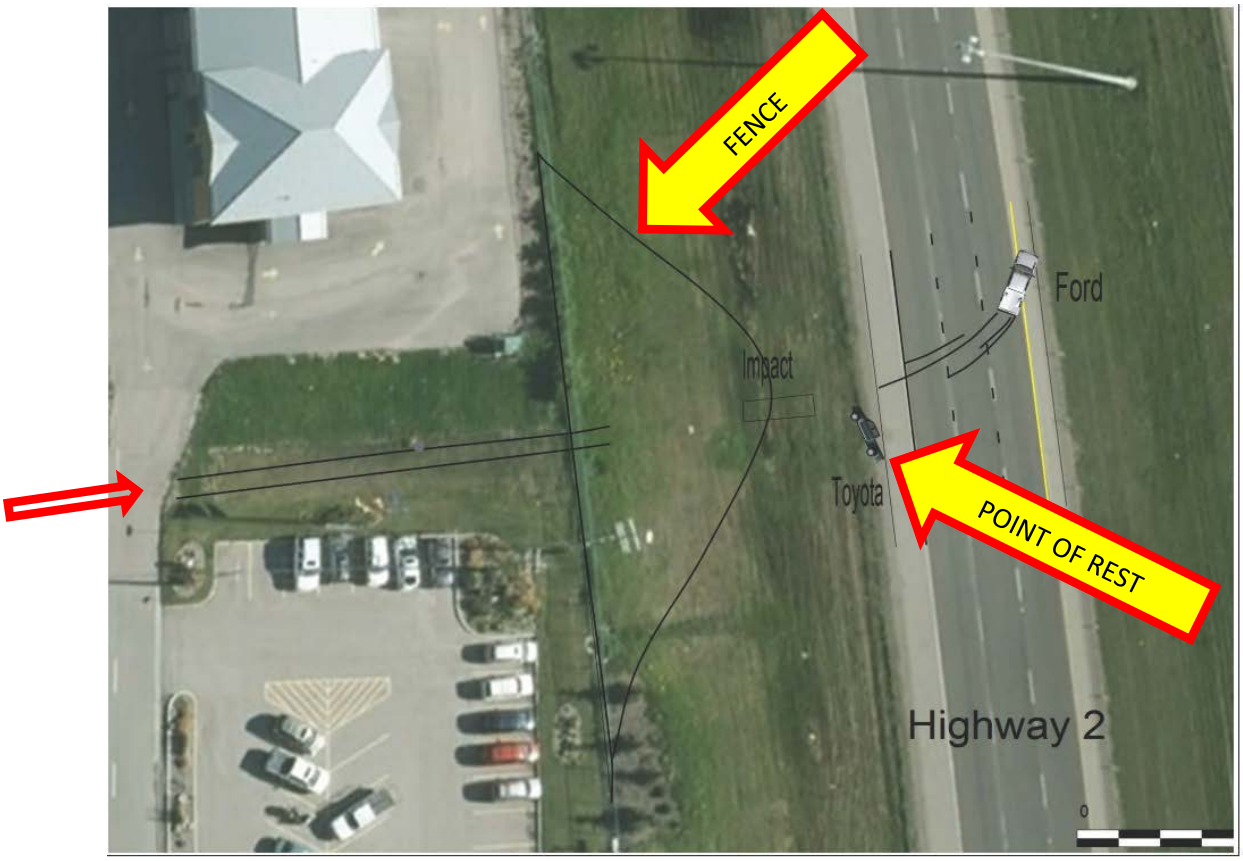
A perpetrator steals a 2014 Toyota Tacoma pickup (as his buddy steals an adjoining 2013 F150) and is running from police at a high rate of speed when he gets to a T intersection with no road ahead. The driver was a long way ahead of the police and there were no witnesses to the night-time crash. He hits a curb, then a fence, then tumbles down a hill to the edge of a limited access highway. The driver fled on foot and was later apprehended. For training purposes, assume the driver had a passenger that was killed in the crash and this is a traffic homicide investigation. (There was no passenger, this scenario is for training).

The purpose of the case is

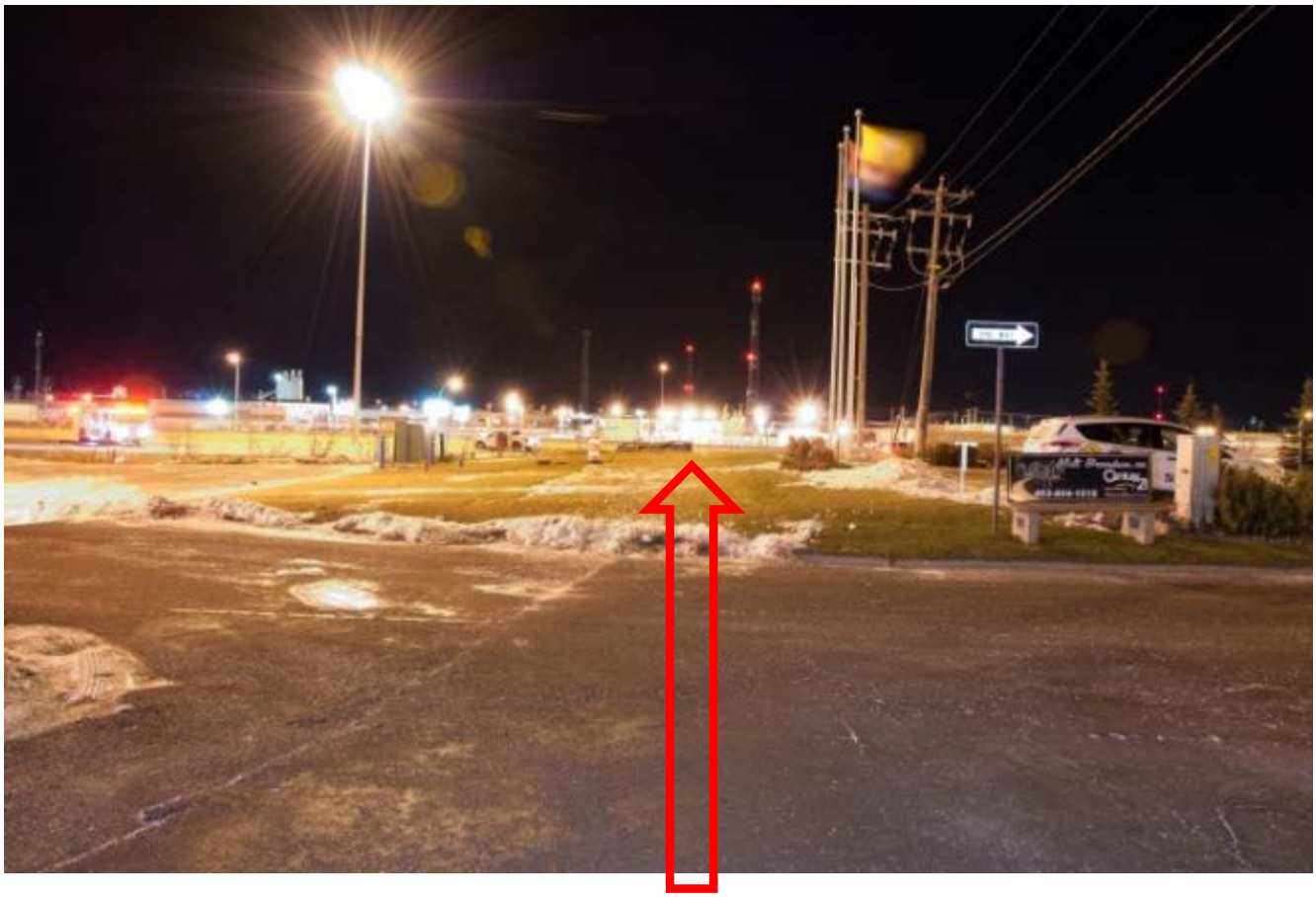
1) to expose the EDR student to the new “13 EDR” file contents which have lateral acceleration in m/sec2 but no Delta V calculated for them
2) to expose the student to a file with multiple events and to tie the multiple events to locations on the scene diagram
3) to expose the student to the need to use Microsoft excel to calculate lateral Delta V from lateral acceleration data, and roll angle from roll rate data.

Review the scene photographs and the attached EDR file. State what EDR events correspond to what physical events in your scene diagram.




FENCE
Ford
Impact
Toyota
POINT OF REST
Highway 2
0


Edmonton Trail NE
2002
Imagery Date: 6/3/2013
Google earth



http://www.1010tires.com/Tools/Tire-Size-Calculator/265-60R18/265-65R17

File Edit View Favorites Tools Help

Google Search Share More

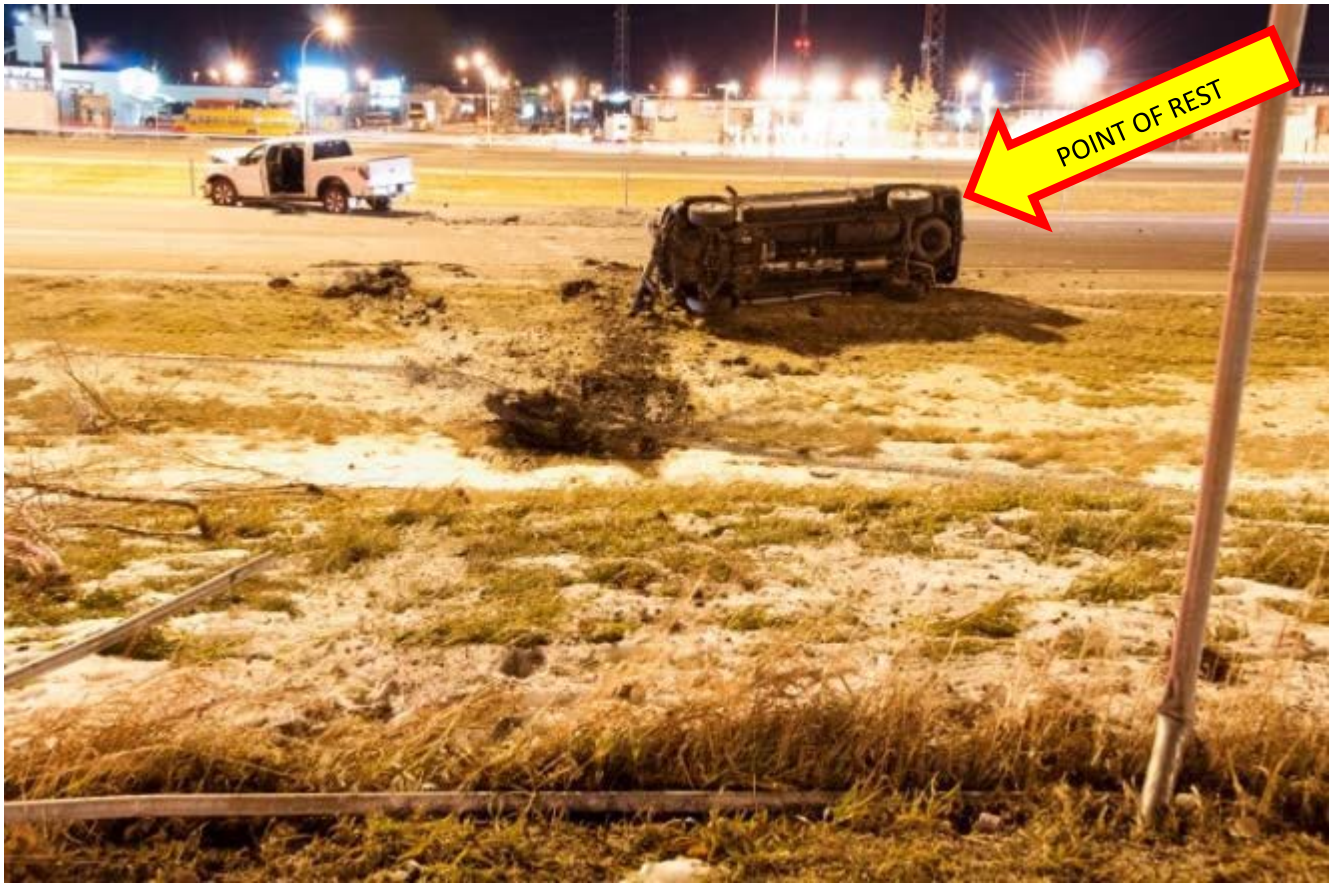
Stock Tire Size - 265/60R18

	Imperial	Metric
Section Width:	10.43 in	265.00 mm
Rim Diameter:	18 in	457.2 mm
Rim Width Range:	7.5 - 9.5 in	
Overall Diameter:	30.52 in	775.20 mm
Sidewall Height:	6.26 in	159.00 mm
Radius:	15.26 in	387.60 mm
Circumference:	95.88 in	2,435.36 mm
Revs per Mile:	681.53/mi	
Actual Speed:	62.14 mph	100.00 kph
Speed Variance:	-	
Diameter Variance:	-	

Tire Size 1 - 265/65R17

	Imperial	Metric
Section Width:	10.43 in	265.00 mm
Rim Diameter:	17 in	431.8 mm
Rim Width Range:	7.5 - 9 in	
Overall Diameter:	30.56 in	776.30 mm
Sidewall Height:	6.78 in	172.25 mm
Radius:	15.28 in	388.15 mm
Circumference:	96.02 in	2,438.82 mm
Revs per Mile:	680.56/mi	
Actual Speed:	62.23 mph	100.14 kph
Speed Variance:	0.14% too slow.	
Diameter Variance:	0.14%	


POINT OF REST
FENCE


POINT OF REST

CASE STUDY QUESTIONS – STUDENT COPY

1) How many recorded events are there in the life of this vehicle, and of the recorded events how many are related to this crash?? Summarize the events in the chart below. Not all data elements are available for all crash types, fill in what you can and say N.A. for not available. Only put in numbers from the EDR report, do not do calculations yet.

Event	Type	Direction	What Deployed?	First Speed	Last Speed	Long DV	Lat DV or Max Accel	Max Roll Angle
6	Side	Pass	ND	93.2	43.5	N.A.	86.2 m/sec2	N. A.
5	Front	Front						
4	Front							
3	Roll							
2	Side							
1	Roll							

2) Identify the physical event that each event related to your crash corresponds to

   a. Calculate the lateral Delta V for the Frontal Events using the CSV file on your student reference disks under the projects tab. The lateral Delta V is in meters per second squared. If you work in English units, divide by gravity 9.8 m/sec2 to get g's, multiply by 32.2/1.466 to get mph/sec, multiply by .004 seconds time to get Delta V in mph for that interval. Add up the 25 intervals. Save the file as a Microsoft excel file type.

   b. Calculate Total Delta V and PDOF for the frontal events.

3) What was the fastest speed recorded anytime in the 10 seconds prior to impact?

4) If the speedometer is right, the EDR is right. If the speedometer is wrong, the EDR is wrong. The tires on the vehicle are 265/60R18, the factory tires were 265/65R17. What adjustment factor is needed to correct the speedometer and the EDR readings? Using the adjustment factor, what is the fastest nominal speed now?

5) If you were filing criminal charges, what speed could you say beyond a reasonable doubt that the driver was going at least that fast?

6) What is the highest brake pressure the driver applied in the 5 seconds before the last crash? What does that mean in terms of braking percentage?

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN/Frame Number	5TFMU4FN6EX******
User	portz
Case Number	14-1482783_toyota
EDR Data Imaging Date	11/25/2014
Crash Date	11/25/2014
Filename	2014 TACOMA SINGLE VEHICLE CRASH.CDRX
Saved on	Tuesday, November 25 2014 at 16:40:50
Collected with CDR version	Crash Data Retrieval Tool 14.2
Reported with CDR version	Crash Data Retrieval Tool 16.4
EDR Device Type	Airbag Control Module
Event(s) recovered	Front/Rear (2), Side (2), Rollover (2)

Comments

dlc - reconnect cut battery

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Pre-Crash data is recorded in discrete intervals. Due to different refresh rates within the vehicle's electronics, the data recorded may not be synchronous to each other.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If any of the front passenger seat airbags, side airbags, or Curtain Shield Airbags have deployed, data will not be overwritten or deleted by the airbag ECU following that event. If none of the airbags have deployed, the data of that event may be overwritten by a following event even if other airbags (pretensioner, rear seat airbag, etc.) have deployed.
- If power supply to the airbag ECU is lost during an event, all or part of the data may not be recorded.
- "Diagnostic Trouble Codes" are information about faults when a recording trigger is established. Various diagnostic trouble codes could be set and recorded due to component or system damage during an accident.
- The airbag ECU records only diagnostic information related to the airbag system. It does not record diagnostic information related to other vehicle systems.
- The TaSCAN, Global TechStream, or Intelligent Tester II devices (or any other Toyota genuine diagnostic tool) can be used to obtain detailed information on the diagnostic trouble codes from the airbag system, as well as diagnostic information from other systems. However, in some cases, the diagnostic trouble codes of the airbag system recorded by the airbag ECU when the event occurred may not match the diagnostic trouble codes read out when the diagnostic tool is used.

General Information:

- The data recording specifications of Toyota's airbag ECUs are divided into the following eight categories. The specifications for 12EDR or later are designed to be compatible with NHTSA's 49CFR Part 563 rule.
  - 00EDR / 02EDR / 04EDR / 06EDR / 10EDR / 12EDR / 13EDR / 15EDR
- The airbag ECU records data for all or some of the following accident types: frontal crash, rear crash , side crash, and rollover events. Depending on the installed airbag ECU, data for side crash and/or rollover events may not be recorded.
- This airbag ECU records record pre-crash data and post-crash data.
  - If a single event occurs independently, the data for that event is recorded on a one-to-one basis.
  - If multiple events occur successively (within a period of approximately 500ms), the establishment of the recording trigger for the first event is defined as the "pre-crash recording trigger". Pre-crash data for the first event and post-crash data for each successive event is then recorded.
- The airbag ECU has two recording pages (memory maps) to store pre-crash data. Additionally, to store post-crash data, the airbag ECU has two recording pages for each accident type: two pages for frontal and rear crash, two pages for a side crash, and two pages for rollover event.
- The data recorded by the airbag ECU includes correlating information between each previously occurring event (i.e., information that clarifies the collision event sequence. This correlation information consists of the following items.
  - Time from Previous Pre-Crash TRG
  - Linked Pre-Crash Page
  - Time from Pre-Crash TRG
  - TRG Count
  - Previous Crash Type
- In frontal and rear collision events, the first point where a longitudinal cumulative delta-V of over 0.8 km/h (0.5 mph) is reached is regarded as time zero for the recorded data. In side impact collision and rollover events, the point in time at which the recording trigger is established

is regarded as time zero for the recorded data.

- The recording trigger judgment threshold value differs depending on the collision type (i.e., frontal crash, rear crash, side crash, or rollover event).
- Some of the data recorded by the airbag ECU is transmitted to the airbag ECU from various vehicle control modules by the vehicle's Controller Area Network (CAN).
- In some cases, the airbag ECU part number printed on the ECU label may not match the airbag ECU part number that the CDR tool reports. The part number retrieved by the CDR tool should be considered as the official ECU part number.
- In frontal and rear collision events, the record time varies depending on the period during which a longitudinal cumulative delta-V of over 0.8 km/h (0.5 mph) is reached, and time series data is recorded for up to 250 ms. The record time described above is indicated as "Length of Delta-V". "Delta-V, Longitudinal" outside the record time is indicated by area shaded in the table, and not indicated in the graph.

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report.

Data Element Name	Positive Sign Notation Indicates
Maximum Delta-V, Longitudinal	Forward
Delta-V, Longitudinal	Forward
Lateral Acceleration for Frontal/Rear Crash, Floor Sensor	Left to Right
Lateral Acceleration, Side Satellite Sensor 1	Left to Right
Lateral Acceleration, Side Satellite Sensor 2	Left to Right
Lateral Acceleration, Side Satellite Sensor 3	Left to Right
Lateral Acceleration, Side Satellite Sensor 4	Left to Right
Lateral Acceleration for Side Crash, Floor Sensor	Left to Right
Roll Angle Peak	Clockwise Rotation
Roll Angle at the Time of TRG	Clockwise Rotation
Roll Rate	Clockwise Rotation
Lateral Acceleration for Rollover, Floor Sensor	Left to Right
Longitudinal Acceleration , VSC Sensor	Forward
Yaw Rate	Left Turn
Steering Input	Left Turn

Data Definitions:
1)

- The "ON" setting for the "Freeze Signal" indicates a state in which the non-volatile memory can not be overwritten or deleted by the airbag ECU. After "Freeze Signal" has been turned ON, subsequent events will not be recorded.
- "Recording Status" indicates a state in which all recorded event data has been written into the non-volatile memory, or a state in which this process was interrupted and not fully written into the non-volatile memory. If "Recording Status" is "Incomplete", recorded event data may not be valid.
- If the "Occupant Size Classification, Front Passenger" displays "Child" or "Not Occupied", "Side Air Bag Deployment, Time to Deploy" and "Pretensioner Deployment, Time to Fire" may indicate a time even if deployment did not occur on the for following part no's:
  - 89170-07280, 35400, 35410, 35470, 42660, 0R120, 0R080, 0R081, 0R150
- "Engine RPM" indicates the number of engine revolutions, not the number of motor revolutions. The recorded value has an upper limit of 12,800 rpm. Resolution is 100 rpm and the value is rounded down and recorded. For example, if the actual engine speed is 799 rpm, the recorded value will be 700 rpm.
- If the electric vehicle is using a calculated/virtual engine RPM for drivetrain control, "Engine RPM" may be recorded, but should not be used during data analysis.
- The upper limit for the recorded "Vehicle Speed" value is 200 km/h (125mph). Resolution is 1km/h (0.6mph) and the value is rounded down and recorded. The accuracy of the "Vehicle Speed" value can be affected by various factors. These include, but not limited, to the following.
  - Significant changes in the tire's rolling radius
  - Wheel lock and wheel slip
- "Accelerator Pedal" has two recording specifications. Both the recorded value increases as the driver depresses the accelerator.
  - Percentage of accelerator pedal depressed (recorded as 0-100(%)).
  - Output voltage of accelerator pedal module (recorded as 0-5(V)).
- If M/T transmission vehicle of some limited model, "Shift Position" may display "Drive" regardless of the actual shift position.
- Depending on the type of occupant sensor installed in the vehicle, one of the following three recording formats for "Occupant Size Classification, Front Passenger" will be utilized.
  - Occupied / Not Occupied
  - AM50 / AF05 / Child / Not Occupied
  - AM50 / AF05 / Child or Not Occupied
- "Cruise Control Status" indicates whether the cruise control system is actuated or not. OFF indicates that the cruise control system is not actuated, but can also indicates that the vehicle is not equipped with the system.
- "Air Bag Warning Lamp, On/Off", "Ignition Cycle, Crash", "Seat Track Position Switch, Foremost, Status, Driver", "Occupant Size Classification, Front Passenger", "Safety Belt Status, Driver", "Safety Belt Status, Front Passenger", "Frontal Air Bag Suppression Switch Status, Front Passenger", and "RSCA Disable Switch" indicate the state approximately 1 second before time zero. They may not always indicate the state at the moment of collision.
- The upper and lower limits for the recorded value of "Motor RPM" is 17,500 rpm and -7,500 rpm respectively. Resolution is 100 rpm and the value is rounded down and recorded.
- "Brake Oil Pressure" has an upper limit of 12.14 Mpa. In the case of the vehicle that has not VSC system, "0 Mpa" or "Invalid" may be displayed.

- "Longitudinal Acceleration , VSC Sensor" has upper and lower limits for the recorded value of 8.973 m/s^2 and -8.973 m/s^2 respectively. This acceleration sensor does not sense collisions.
- "Sequential Shift Range" displaying "Undetermined" indicates the shift range is undetermined or was not being used.
- Some vehicles will not be equipped with all "Drive Mode" types indicated in the "Drive Mode" table. If some or all drive modes are not applicable to vehicle, "OFF" or "Invalid" may be displayed. The item in the "Drive Mode" table may not match the name of switch or indicator that equipped the vehicle.
- The upper and lower limits for the recorded value of "Steering Input" is 375 deg and -375 deg respectively. Resolution is 1.5 deg and the value is rounded down and recorded.
- Resolution of the "Air Bag Warning Lamp ON Time Since DTC was Set" is 15 minutes, and the value is rounded down and recorded.
- "Delta-V, Longitudinal" indicates the change in forward speed after time zero. This does not refer to vehicle speed, and it does not include the change in speed during the period from the start of the actual collision to establishment of the time zero.
- "Location of Side Satellite Sensor" shows the outline of a typical sensor position. Sensory location can be confirmed using the repair manual.
- "Time from Previous Pre-Crash TRG" indicates the time between the establishment of an event's pre-crash recording trigger to the establishment of a more recent event's pre-crash recording trigger. The upper limit for the recorded value is 16,381 milliseconds. In the event of establishment of the first pre-crash recording trigger after the ignition is switched ON, the upper limit value(max value) is recorded.
- "TRG Count" indicates a calculated value of the number of times recording triggers have been established for all crash types. The sequence in which each event occurred can be verified from the "TRG Count". The smaller the "TRG Count" value, the older the data. The upper limit for the recorded value is 65,533 times. When more than one event reaches the upper limit, the actual "TRG Count" may be greater than what is displayed for that event.
- "Linked Pre-Crash Page" is used to link 'paged" pre-crash data with 'paged" post-crash data. When old pre-crash data is overwritten by new pre-crash data, the "Linked Pre-Crash Page" value may record a page number that is not actually linked.
- Resolution of the "Time from Pre-Crash to TRG" is 50 [ms], and the value is rounded up and recorded.
- "Roll Angle at the Time of TRG" and "Roll Angle Peak" do not represent the actual roll angle of the vehicle. These values are used internally by the airbag ECU for sensing a rollover.

05013_ToyotaS00std_r025

System Status at Time of Retrieval

ECU Part Number	89170-04590
EDR Generation	13EDR
Complete File Recorded	Yes
Freeze Signal	ON
Freeze Signal Factor	Rollover CSA Deployment
Diagnostic Trouble Codes Exist	No
Ignition Cycle ,Download (times)	713
Multi-event, number of events (times)	2 or greater
Time from event 1 to 2 (s)	0.059
Time from Previous Pre Crash TRG (msec)	522
Latest Pre-Crash Page	1
Contains Unlinked Pre-Crash Data	No

Event Record Summary at Retrieval

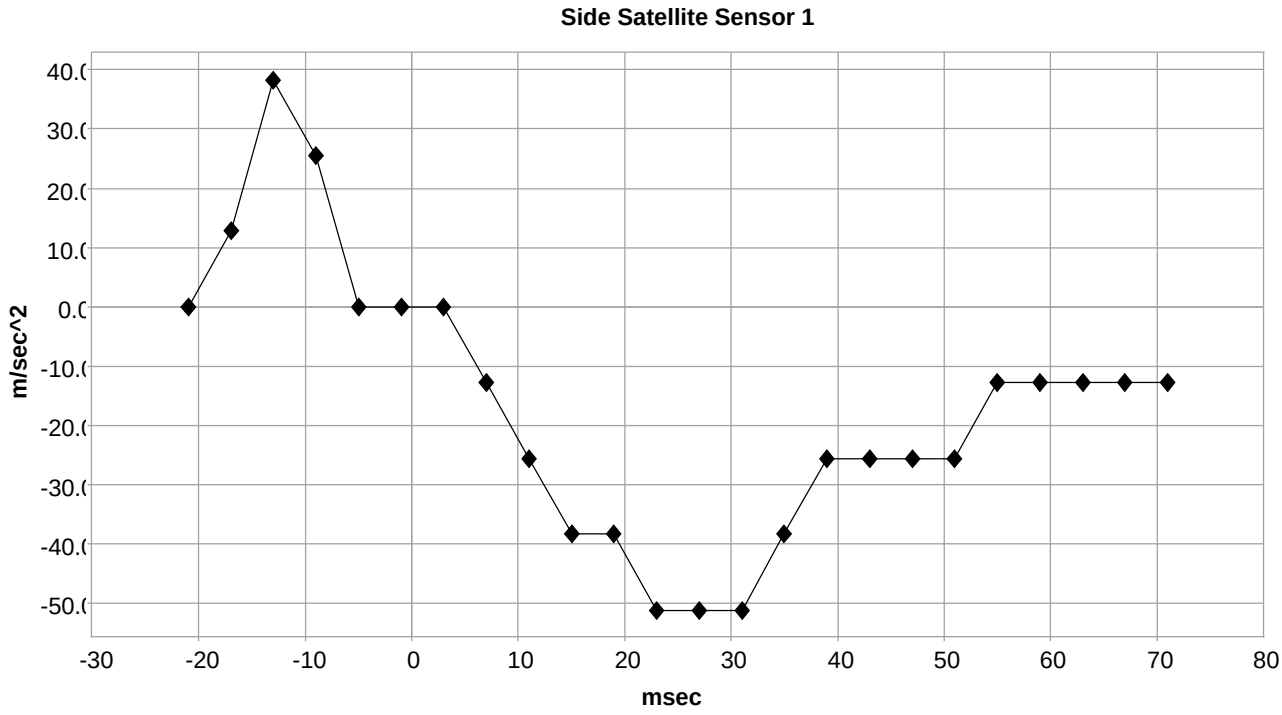
Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	6	Side Crash	0	Complete (Page 1)	Complete (Side Page 1)
1st Prior Event	5	Front/Rear Crash	-59	Complete (Page 1)	Complete (Front/Rear Page 1)
2nd Prior Event	4	Front/Rear Crash	-687	Complete (Page 0)	Complete (Front/Rear Page 0)
3rd Prior Event	3	Rollover	N/A	N/A	Complete (Rollover Page 1)
4th Prior Event	2	Side Crash	N/A	N/A	Complete (Side Page 0)
5th Prior Event	1	Rollover	N/A	N/A	Complete (Rollover Page 0)

System Status at Event (Most Recent Event, TRG 6)

Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	6
Previous Crash Type	Front/Rear Crash
Time from Pre-Crash TRG (msec)	18
Linked Pre-Crash Page	1
Side Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Side Curtain Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Pretensioner Deployment, Time to Fire (msec)	SNA
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

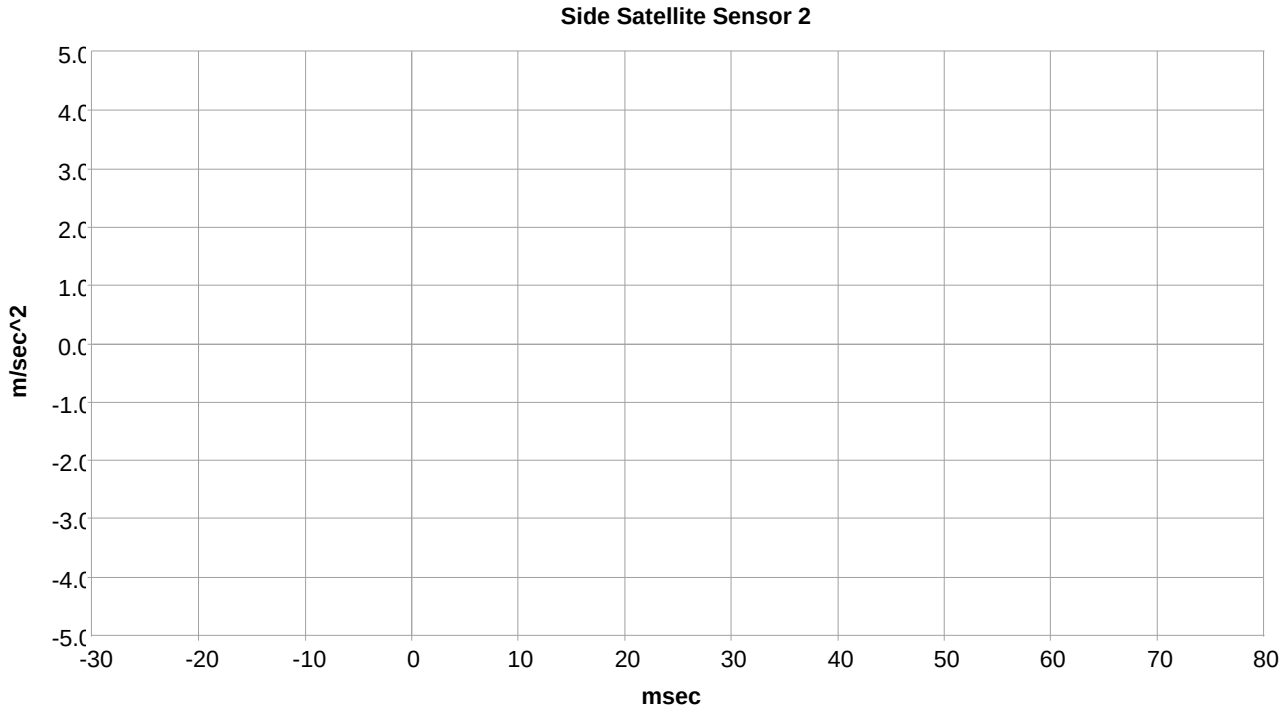
Lateral Crash Pulse (Most Recent Event, TRG 6 - table 1 of 2)

Recording Status, Time Series Data	Complete
Recorded Side	Right Side
Time from TRG to Next Sample (msec)	3
Location of Side Satellite Sensor 1	Front Rocker
Location of Side Satellite Sensor 2	Not Equipped
Location of Side Satellite Sensor 3	C-Pillar
Location of Side Satellite Sensor 4	Not Equipped
Location of Floor Sensor	Airbag ECU
Clipping Time of Lateral Acceleration, Side Satellite Sensor 1 (msec)	No
Clipping Time of Lateral Acceleration, Side Satellite Sensor 2 (msec)	SNA
Clipping Time of Lateral Acceleration, Side Satellite Sensor 3 (msec)	No
Clipping Time of Lateral Acceleration, Side Satellite Sensor 4 (msec)	SNA
Clipping Time of Lateral Acceleration, Floor Sensor (msec)	No



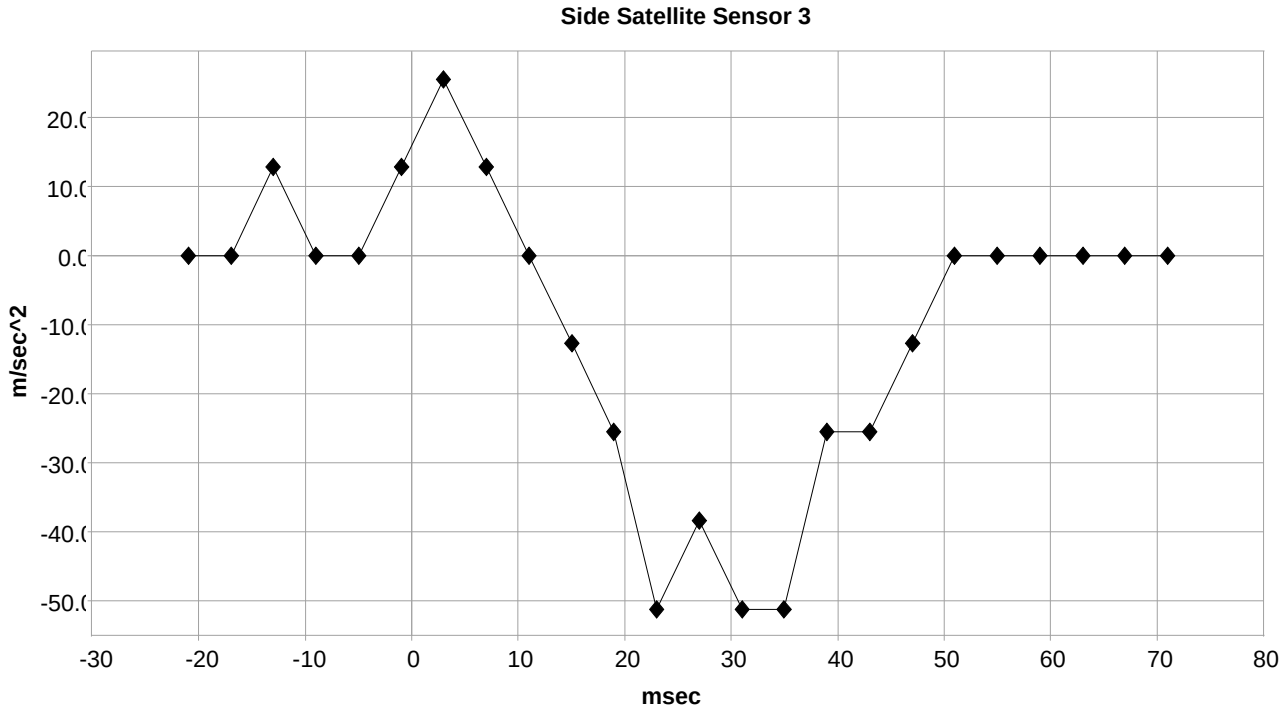
Deployment Time Marker Key

1	Driver/Passenger Pretensioner
2	Side Airbag
3	Rear Window Airbag Deployment Time



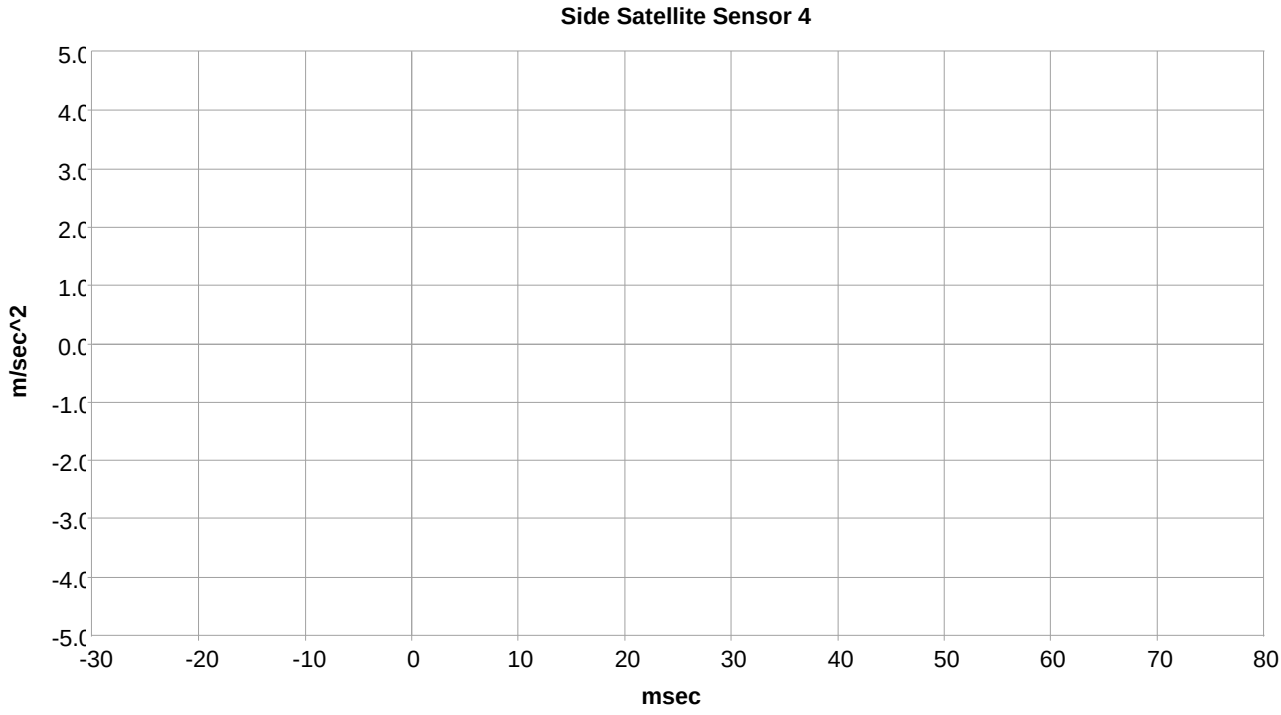
Deployment Time Marker Key

1	Driver/Passenger Pretensioner
2	Side Airbag
3	Rear Window Airbag Deployment Time



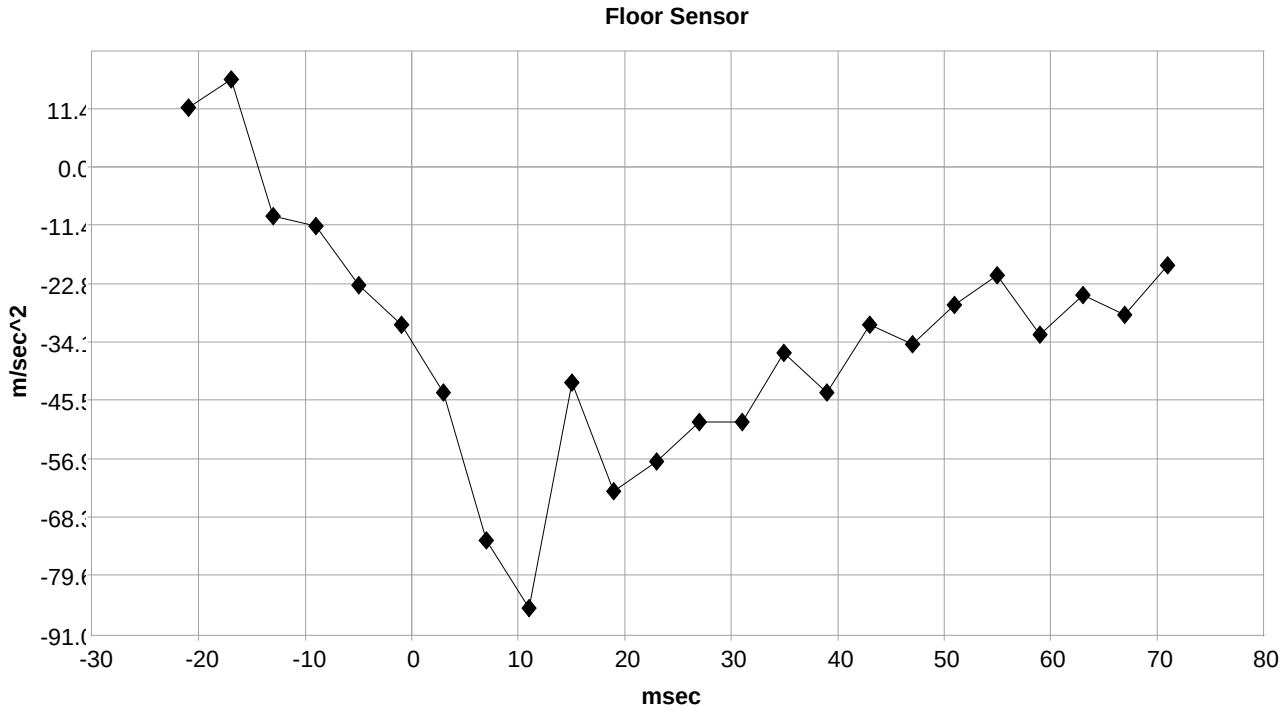
Deployment Time Marker Key

1	Side Curtain Airbag



Deployment Time Marker Key

1	Side Curtain Airbag



Deployment Time Marker Key

1	Driver/Passenger Pretensioner
2	Side Airbag
3	Rear Window Airbag Deployment Time
4	Side Curtain Airbag

Lateral Crash Pulse (Most Recent Event, TRG 6 - table 2 of 2)

Time (msec)	Lateral Acceleration, Side Satellite Sensor 1 (m/sec^2)	Lateral Acceleration, Side Satellite Sensor 2 (m/sec^2)	Lateral Acceleration, Side Satellite Sensor 3 (m/sec^2)	Lateral Acceleration, Side Satellite Sensor 4 (m/sec^2)	Lateral Acceleration for Side Crash, Floor Sensor (m/sec^2)
-21	0.0	SNA	0.0	SNA	11.5
-17	12.8	SNA	0.0	SNA	17.2
-13	38.3	SNA	12.8	SNA	-9.6
-9	25.5	SNA	0.0	SNA	-11.5
-5	0.0	SNA	0.0	SNA	-23.0
-1	0.0	SNA	12.8	SNA	-30.6
3	0.0	SNA	25.5	SNA	-44.1
7	-12.8	SNA	12.8	SNA	-72.8
11	-25.5	SNA	0.0	SNA	-86.2
15	-38.3	SNA	-12.8	SNA	-42.1
19	-38.3	SNA	-25.5	SNA	-63.2
23	-51.1	SNA	-51.1	SNA	-57.5
27	-51.1	SNA	-38.3	SNA	-49.8
31	-51.1	SNA	-51.1	SNA	-49.8
35	-38.3	SNA	-51.1	SNA	-36.4
39	-25.5	SNA	-25.5	SNA	-44.1
43	-25.5	SNA	-25.5	SNA	-30.6
47	-25.5	SNA	-12.8	SNA	-34.5
51	-25.5	SNA	0.0	SNA	-26.8
55	-12.8	SNA	0.0	SNA	-21.1
59	-12.8	SNA	0.0	SNA	-32.6
63	-12.8	SNA	0.0	SNA	-24.9
67	-12.8	SNA	0.0	SNA	-28.7
71	-12.8	SNA	0.0	SNA	-19.2

DTCs Present at Time of Event (Most Recent Event, TRG 6)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent Event, TRG 6)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	500
TRG Count when Pre-crash TRG was Established (times)	5
Safety Belt Status, Driver	ON
Safety Belt Status, Front Passenger	OFF
Occupant Size Classification, Front Passenger	Not Occupied
Frontal Airbag Suppression Switch Status, Front Passenger	SNA
RSCA Disable Switch	OFF (enable)
Seat Track Position Switch, Foremost, Status, Driver	Yes
Airbag Warning Lamp, On/Off	OFF
Ignition Cycle ,Crash (times)	706

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 6)

Time (sec)	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0 (TRG)
Vehicle Speed (MPH [km/h])	93.2 [150]	93.2 [150]	92.6 [149]	91.3 [147]	78.9 [127]	81.4 [131]	77.7 [125]	70.8 [114]	70.8 [114]	58.4 [94]	43.5 [70]
Accelerator Pedal, % Full (%)	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	32.0	42.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Engine RPM (RPM)	4,100	4,100	3,800	3,400	2,700	2,300	2,200	1,900	1,800	1,700	1,200
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	2.64	4.75	2.74	10.22	11.81	12.14	12.14	12.14
Longitudinal Acceleration , VSC Sensor (m/sec^2)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Yaw Rate (deg/sec)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Steering Input (degrees)	3.0	1.5	10.5	15.0	16.5	34.5	27.0	19.5	21.0	30.0	7.5
Shift Position	D	D	D	D	D	D	D	D	D	D	D
Sequential Shift Range	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Cruise Control Status	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, PWR	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, ECO	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, Sport	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, Snow	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

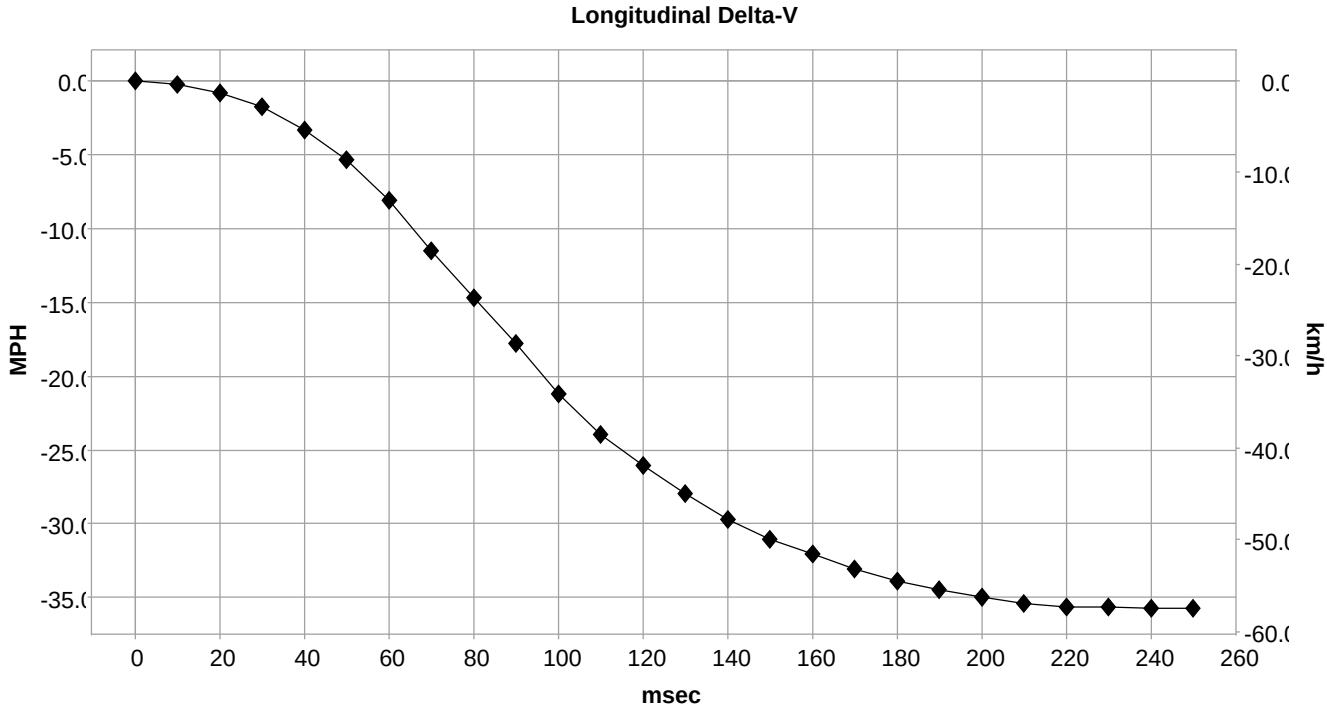
Drive Mode, EV	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Fuel Injection Quantity (mm3/st)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

System Status at Event (1st Prior Event, TRG 5)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	5
Previous Crash Type	Front/Rear Crash
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	1
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (msec)	No
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front Passenger (msec)	No
Pretensioner Deployment, Time to Fire, Driver (msec)	No
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	No
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	N/A
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (msec)	N/A
Active Head Restraint, Time to Deploy, Driver (msec)	SNA
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec)	SNA
Side Airbag Deployment, Time to Deploy, Driver (msec)	SNA
Side Airbag Deployment, Time to Deploy, Passenger (msec)	SNA
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

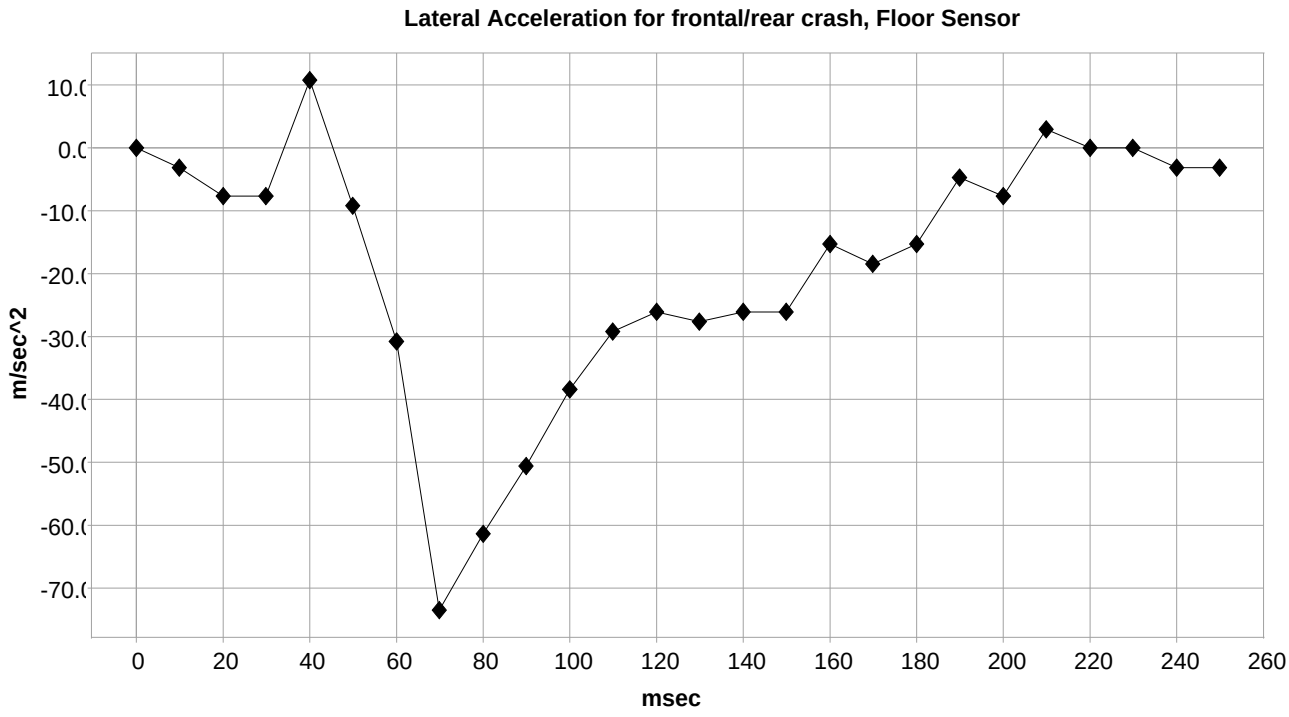
Longitudinal/Lateral Crash Pulse (1st Prior Event, TRG 5 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	41.0
Length of Delta-V (msec)	250
Max. Longitudinal Delta-V (MPH [km/h])	-35.7 [-57.5]
Time, Maximum Delta-V, Longitudinal (msec)	251.0
Power Supply Status at Max. Delta-V	ON
Clipping Time of Longitudinal Delta-V (msec)	No
Clipping Time of Lateral Acceleration, Floor Sensor (msec)	No



Deployment Time Marker Key

1	Driver Airbag Deployment Time
2	Passenger Airbag Deployment Time
3	Driver/Passenger Pretensioner
4	Driver 2nd Stage Airbag Deployment Time
5	Passenger 2nd Stage Airbag Deployment
6	Driver/Passenger AHR
7	Driver CSA
8	Passenger CSA
9	Rear Window Airbag Deployment Time
10	Driver SAB
11	Passenger SAB



Deployment Time Marker Key

1	Driver Airbag Deployment Time
2	Passenger Airbag Deployment Time
3	Driver/Passenger Pretensioner
4	Driver 2nd Stage Airbag Deployment Time
5	Passenger 2nd Stage Airbag Deployment
6	Driver/Passenger AHR
7	Driver CSA
8	Passenger CSA
9	Rear Window Airbag Deployment Time
10	Driver SAB
11	Passenger SAB

Longitudinal/Lateral Crash Pulse (1st Prior Event, TRG 5 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	Power Supply Status
0	0.0 [0.0]	0.0	ON
10	-0.3 [-0.4]	-3.1	ON
20	-0.8 [-1.3]	-7.7	ON
30	-1.7 [-2.8]	-7.7	ON
40	-3.3 [-5.3]	10.7	ON
50	-5.4 [-8.6]	-9.2	ON
60	-8.1 [-13.0]	-30.6	ON
70	-11.5 [-18.6]	-73.5	ON
80	-14.7 [-23.6]	-61.3	ON
90	-17.8 [-28.6]	-50.6	ON
100	-21.2 [-34.2]	-38.3	ON
110	-24.0 [-38.6]	-29.1	ON
120	-26.0 [-41.9]	-26.0	ON
130	-28.0 [-45.0]	-27.6	ON
140	-29.7 [-47.9]	-26.0	ON
150	-31.1 [-50.0]	-26.0	ON
160	-32.1 [-51.6]	-15.3	ON
170	-33.1 [-53.2]	-18.4	ON
180	-33.9 [-54.5]	-15.3	ON
190	-34.5 [-55.5]	-4.6	ON
200	-35.0 [-56.3]	-7.7	ON
210	-35.3 [-56.9]	3.1	ON
220	-35.6 [-57.3]	0.0	ON
230	-35.6 [-57.4]	0.0	ON
240	-35.7 [-57.4]	-3.1	ON
250	-35.7 [-57.4]	-3.1	ON

DTCs Present at Time of Event (1st Prior Event, TRG 5)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (1st Prior Event, TRG 5)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	500
TRG Count when Pre-crash TRG was Established (times)	5
Safety Belt Status, Driver	ON
Safety Belt Status, Front Passenger	OFF
Occupant Size Classification, Front Passenger	Not Occupied
Frontal Airbag Suppression Switch Status, Front Passenger	SNA
RSCA Disable Switch	OFF (enable)
Seat Track Position Switch, Foremost, Status, Driver	Yes
Airbag Warning Lamp, On/Off	OFF
Ignition Cycle ,Crash (times)	706

Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 5)

Time (sec)	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0 (TRG)
Vehicle Speed (MPH [km/h])	93.2 [150]	93.2 [150]	92.6 [149]	91.3 [147]	78.9 [127]	81.4 [131]	77.7 [125]	70.8 [114]	70.8 [114]	58.4 [94]	43.5 [70]
Accelerator Pedal, % Full (%)	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	32.0	42.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Engine RPM (RPM)	4,100	4,100	3,800	3,400	2,700	2,300	2,200	1,900	1,800	1,700	1,200
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	2.64	4.75	2.74	10.22	11.81	12.14	12.14	12.14
Longitudinal Acceleration , VSC Sensor (m/sec^2)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Yaw Rate (deg/sec)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Steering Input (degrees)	3.0	1.5	10.5	15.0	16.5	34.5	27.0	19.5	21.0	30.0	7.5
Shift Position	D	D	D	D	D	D	D	D	D	D	D
Sequential Shift Range	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Cruise Control Status	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, PWR	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, ECO	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, Sport	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, Snow	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

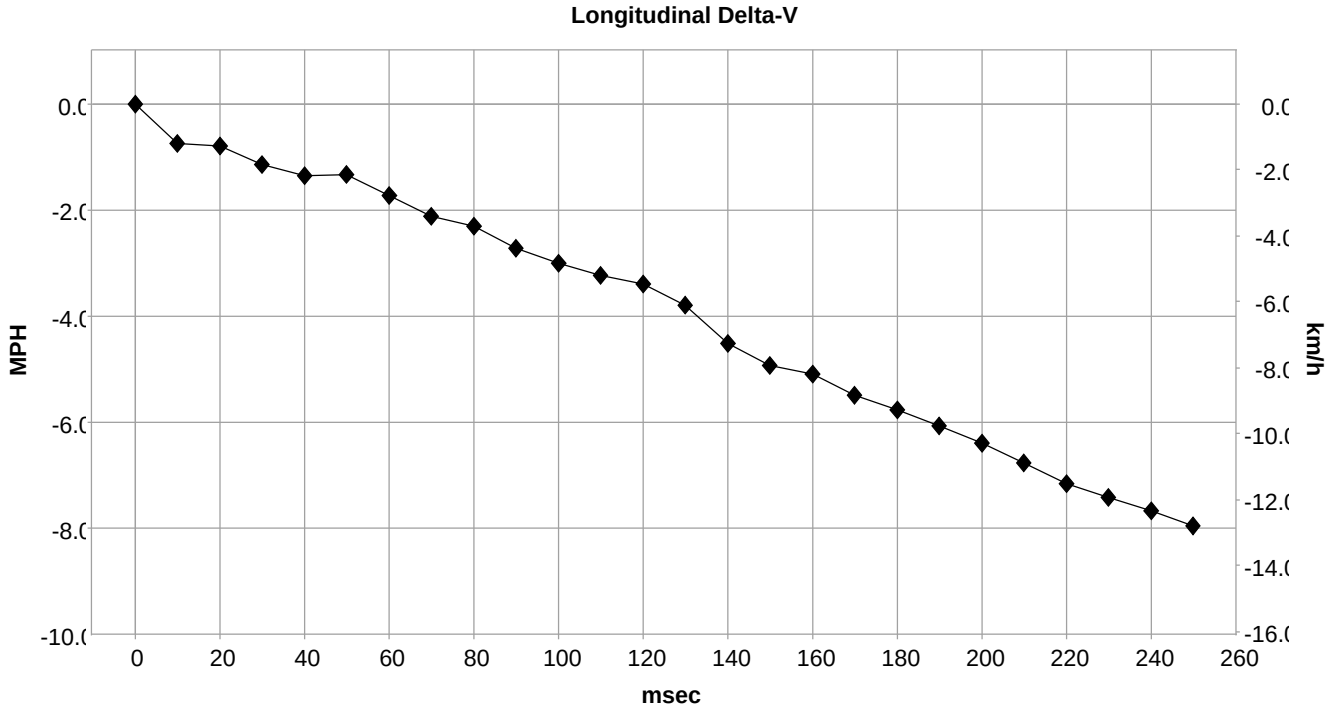
Drive Mode, EV	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Fuel Injection Quantity (mm3/st)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

System Status at Event (2nd Prior Event, TRG 4)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	4
Previous Crash Type	Rollover
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (msec)	No
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front Passenger (msec)	No
Pretensioner Deployment, Time to Fire, Driver (msec)	No
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	No
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	N/A
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (msec)	N/A
Active Head Restraint, Time to Deploy, Driver (msec)	SNA
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec)	SNA
Side Airbag Deployment, Time to Deploy, Driver (msec)	SNA
Side Airbag Deployment, Time to Deploy, Passenger (msec)	SNA
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

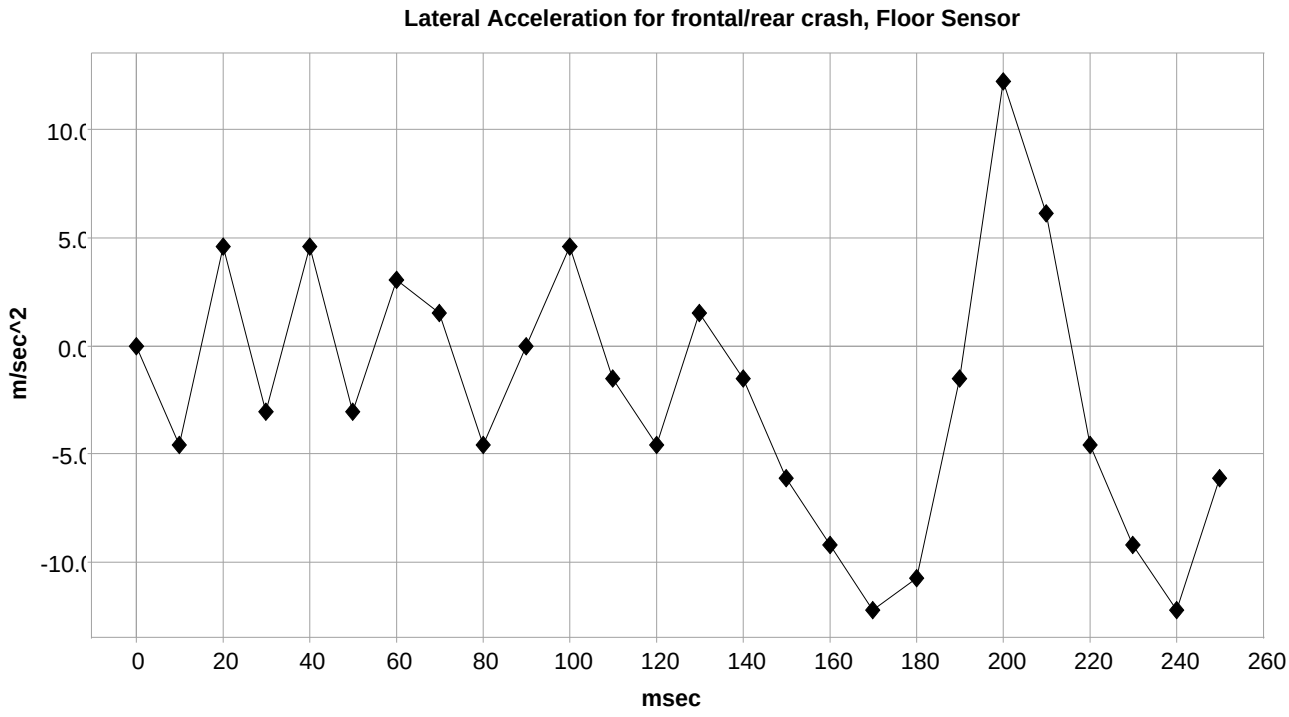
Longitudinal/Lateral Crash Pulse (2nd Prior Event, TRG 4 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	147.0
Length of Delta-V (msec)	250
Max. Longitudinal Delta-V (MPH [km/h])	-9.2 [-14.8]
Time, Maximum Delta-V, Longitudinal (msec)	300.0
Power Supply Status at Max. Delta-V	ON
Clipping Time of Longitudinal Delta-V (msec)	No
Clipping Time of Lateral Acceleration, Floor Sensor (msec)	No



Deployment Time Marker Key

1	Driver Airbag Deployment Time
2	Passenger Airbag Deployment Time
3	Driver/Passenger Pretensioner
4	Driver 2nd Stage Airbag Deployment Time
5	Passenger 2nd Stage Airbag Deployment
6	Driver/Passenger AHR
7	Driver CSA
8	Passenger CSA
9	Rear Window Airbag Deployment Time
10	Driver SAB
11	Passenger SAB



Deployment Time Marker Key

1	Driver Airbag Deployment Time
2	Passenger Airbag Deployment Time
3	Driver/Passenger Pretensioner
4	Driver 2nd Stage Airbag Deployment Time
5	Passenger 2nd Stage Airbag Deployment
6	Driver/Passenger AHR
7	Driver CSA
8	Passenger CSA
9	Rear Window Airbag Deployment Time
10	Driver SAB
11	Passenger SAB

Longitudinal/Lateral Crash Pulse (2nd Prior Event, TRG 4 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	Power Supply Status
0	0.0 [0.0]	0.0	ON
10	-0.7 [-1.2]	-4.6	ON
20	-0.8 [-1.3]	4.6	ON
30	-1.1 [-1.8]	-3.1	ON
40	-1.3 [-2.2]	4.6	ON
50	-1.3 [-2.1]	-3.1	ON
60	-1.7 [-2.8]	3.1	ON
70	-2.1 [-3.4]	1.5	ON
80	-2.3 [-3.7]	-4.6	ON
90	-2.7 [-4.4]	0.0	ON
100	-3.0 [-4.8]	4.6	ON
110	-3.2 [-5.2]	-1.5	ON
120	-3.4 [-5.4]	-4.6	ON
130	-3.8 [-6.1]	1.5	ON
140	-4.5 [-7.3]	-1.5	ON
150	-4.9 [-7.9]	-6.1	ON
160	-5.1 [-8.2]	-9.2	ON
170	-5.5 [-8.8]	-12.3	ON
180	-5.8 [-9.3]	-10.7	ON
190	-6.1 [-9.8]	-1.5	ON
200	-6.4 [-10.3]	12.3	ON
210	-6.8 [-10.9]	6.1	ON
220	-7.2 [-11.5]	-4.6	ON
230	-7.4 [-11.9]	-9.2	ON
240	-7.7 [-12.3]	-12.3	ON
250	-7.9 [-12.8]	-6.1	ON

DTCs Present at Time of Event (2nd Prior Event, TRG 4)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (2nd Prior Event, TRG 4)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	450
TRG Count when Pre-crash TRG was Established (times)	4
Safety Belt Status, Driver	ON
Safety Belt Status, Front Passenger	OFF
Occupant Size Classification, Front Passenger	Not Occupied
Frontal Airbag Suppression Switch Status, Front Passenger	SNA
RSCA Disable Switch	OFF (enable)
Seat Track Position Switch, Foremost, Status, Driver	Yes
Airbag Warning Lamp, On/Off	OFF
Ignition Cycle ,Crash (times)	706

Pre-Crash Data, -5 to 0 seconds (2nd Prior Event, TRG 4)

Time (sec)	-4.95	-4.45	-3.95	-3.45	-2.95	-2.45	-1.95	-1.45	-0.95	-0.45	0 (TRG)
Vehicle Speed (MPH [km/h])	97.6 [157]	93.2 [150]	93.2 [150]	92.6 [149]	91.3 [147]	78.9 [127]	81.4 [131]	77.7 [125]	70.8 [114]	70.8 [114]	64 [103]
Accelerator Pedal, % Full (%)	100.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	99.5	32.0	42.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Engine RPM (RPM)	4,200	4,100	4,100	3,800	3,400	2,700	2,300	2,200	1,900	1,800	1,700
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	ON	ON	ON	ON	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	2.64	4.75	2.74	10.22	11.81	12.14	12.14
Longitudinal Acceleration , VSC Sensor (m/sec^2)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Yaw Rate (deg/sec)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Steering Input (degrees)	7.5	3.0	1.5	10.5	15.0	16.5	34.5	27.0	19.5	21.0	25.5
Shift Position	D	D	D	D	D	D	D	D	D	D	D
Sequential Shift Range	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Cruise Control Status	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, PWR	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, ECO	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, Sport	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Drive Mode, Snow	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

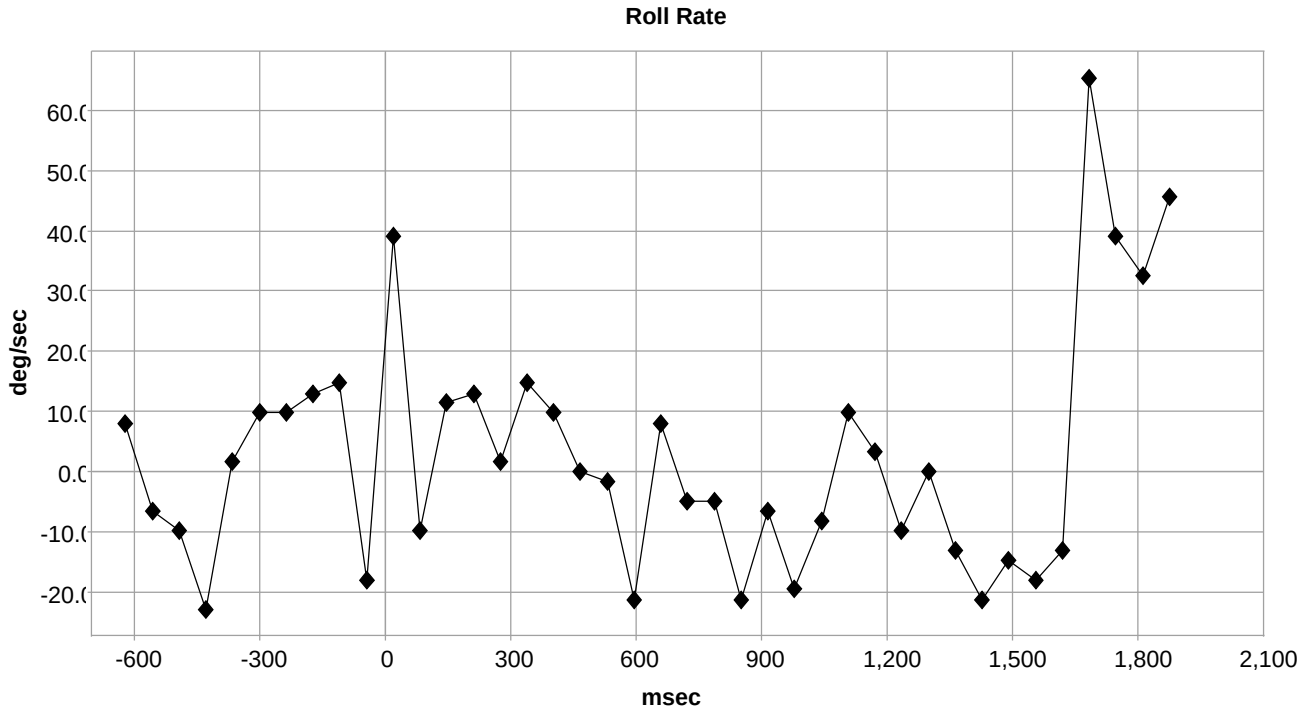
Drive Mode, EV	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Fuel Injection Quantity (mm3/st)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

System Status at Event (3rd Prior Event, TRG 3)

Recording Status, Rollover Crash Info.	Complete
Crash Type	Rollover
TRG Count (times)	3
Previous Crash Type	Side Crash
Time from Pre-Crash TRG (msec)	93
Linked Pre-Crash Page	1
Side Curtain Airbag Deployment, Time to Deploy (msec)	2402
Pretensioner Deployment, Time to Fire, Driver (msec)	2402
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	No

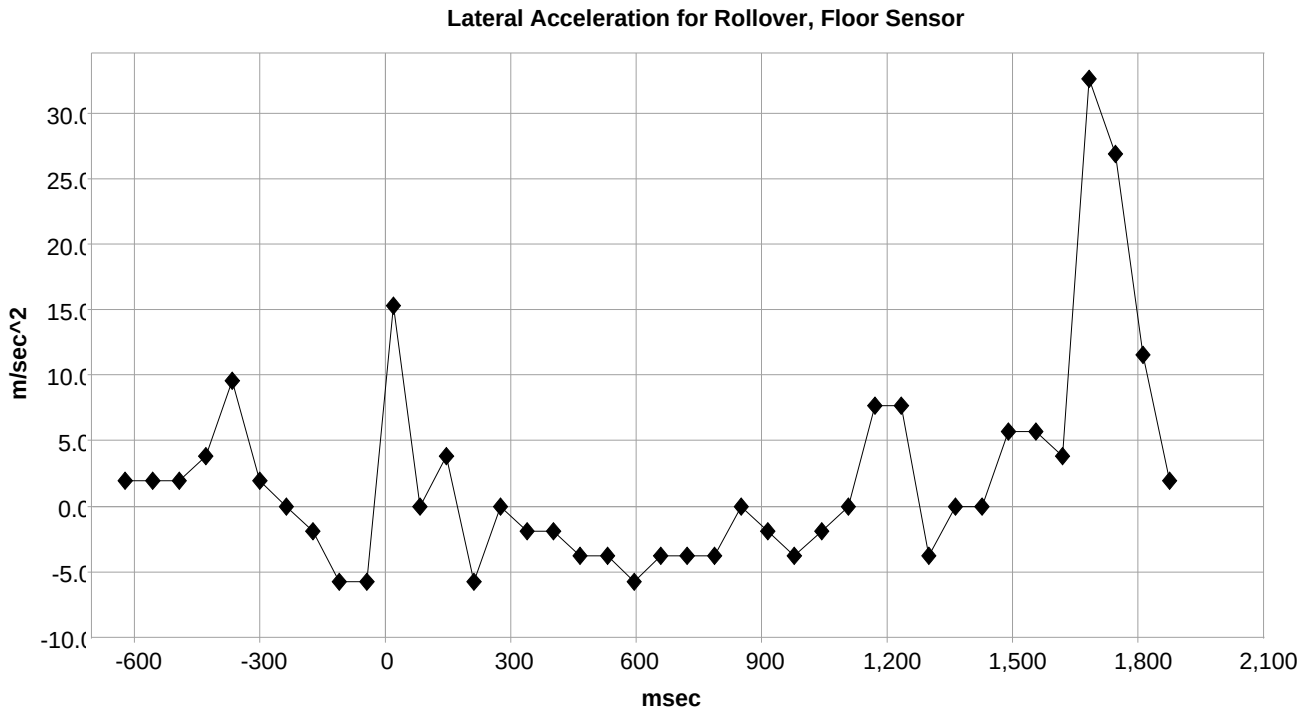
Rollover Crash Pulse (3rd Prior Event, TRG 3 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	19
Roll Angle Peak (degrees)	-57.8
Roll Angle at the Time of TRG (degrees)	3.6



Deployment Time Marker Key

1	Driver/Passenger CSA
2	Driver/Passenger Pretensioner



Deployment Time Marker Key

1	Driver/Passenger CSA
2	Driver/Passenger Pretensioner

Rollover Crash Pulse (3rd Prior Event, TRG 3 - table 2 of 2)

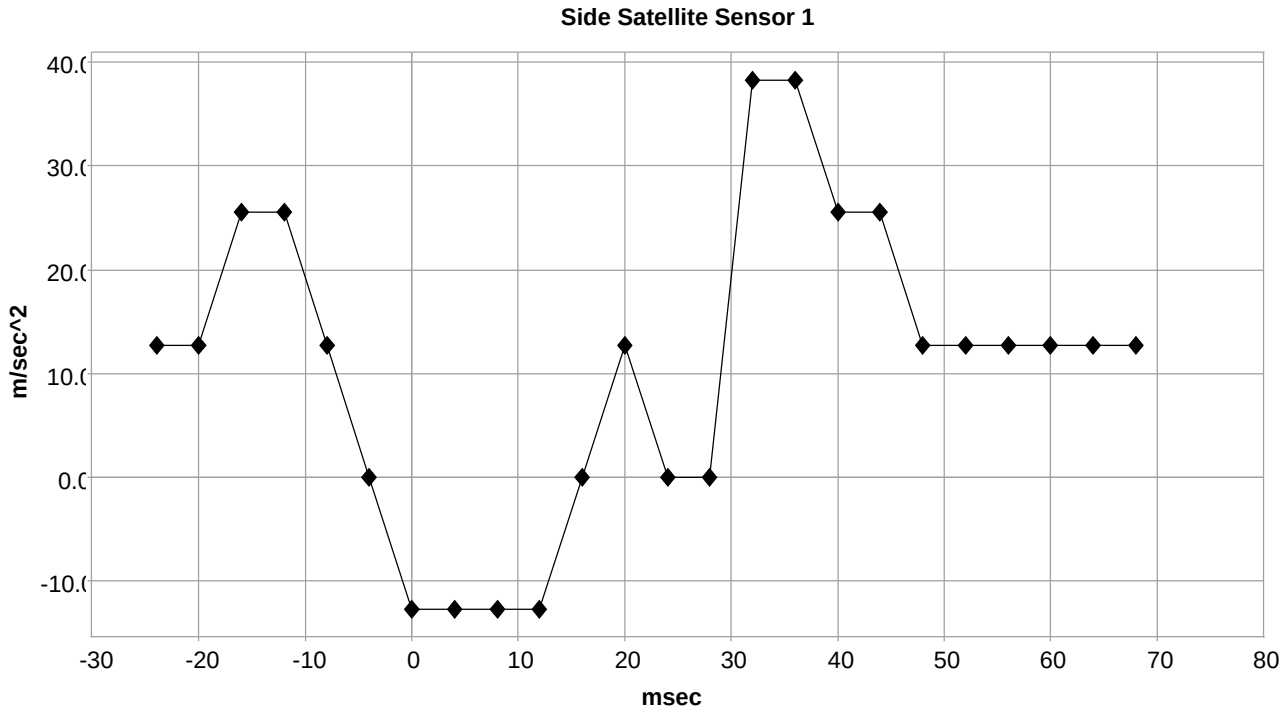
Time (msec)	Roll Rate (deg/sec)	Lateral Acceleration for Rollover, Floor Sensor (m/sec^2)
-621	8.1	1.9
-557	-6.5	1.9
-493	-9.8	1.9
-429	-22.8	3.8
-365	1.6	9.6
-301	9.8	1.9
-237	9.8	0.0
-173	13.0	-1.9
-109	14.7	-5.7
-45	-17.9	-5.7
19	39.1	15.3
83	-9.8	0.0
147	11.4	3.8
211	13.0	-5.7
275	1.6	0.0
339	14.7	-1.9
403	9.8	-1.9
467	0.0	-3.8
531	-1.6	-3.8
595	-21.2	-5.7
659	8.1	-3.8
723	-4.9	-3.8
787	-4.9	-3.8
851	-21.2	0.0
915	-6.5	-1.9
979	-19.5	-3.8
1043	-8.1	-1.9
1107	9.8	0.0
1171	3.3	7.7
1235	-9.8	7.7
1299	0.0	-3.8
1363	-13.0	0.0
1427	-21.2	0.0
1491	-14.7	5.7
1555	-17.9	5.7
1619	-13.0	3.8
1683	65.2	32.6
1747	39.1	26.8
1811	32.6	11.5
1875	45.6	1.9

System Status at Event (4th Prior Event, TRG 2)

Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	2
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	1
Side Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Side Curtain Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Pretensioner Deployment, Time to Fire (msec)	SNA
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

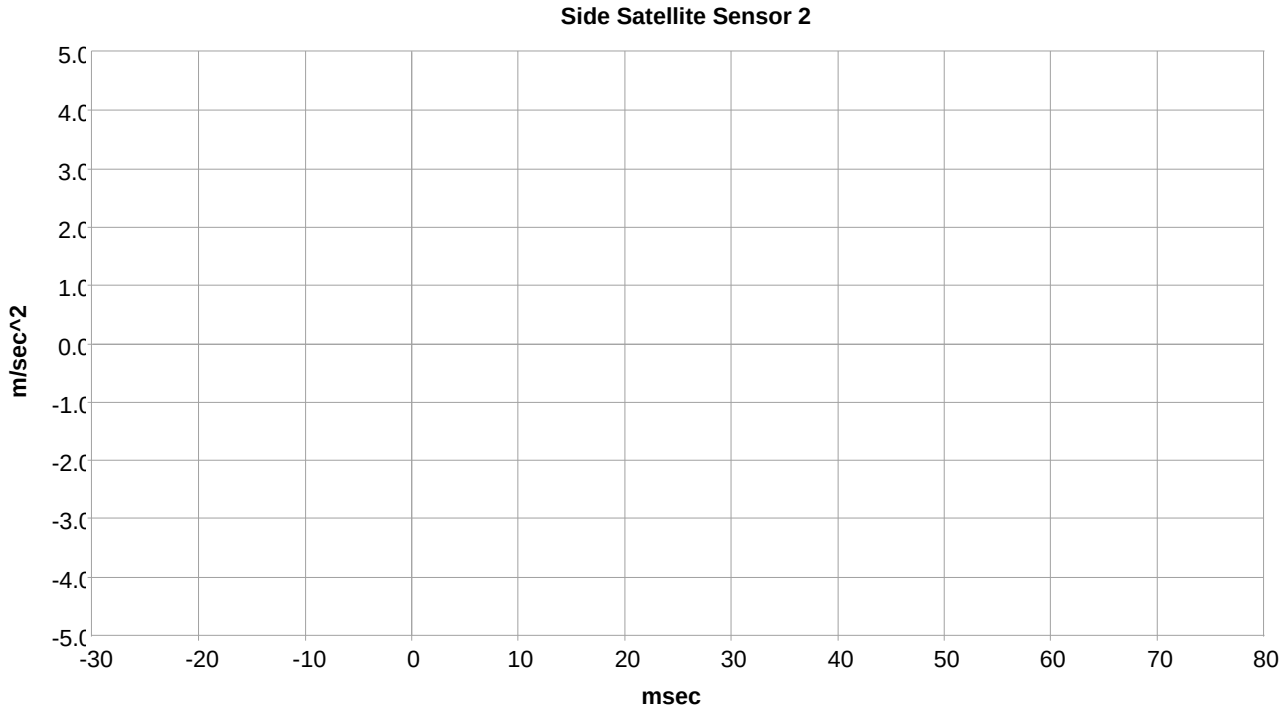
Lateral Crash Pulse (4th Prior Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Recorded Side	Right Side
Time from TRG to Next Sample (msec)	0
Location of Side Satellite Sensor 1	Front Rocker
Location of Side Satellite Sensor 2	Not Equipped
Location of Side Satellite Sensor 3	C-Pillar
Location of Side Satellite Sensor 4	Not Equipped
Location of Floor Sensor	Airbag ECU
Clipping Time of Lateral Acceleration, Side Satellite Sensor 1 (msec)	No
Clipping Time of Lateral Acceleration, Side Satellite Sensor 2 (msec)	SNA
Clipping Time of Lateral Acceleration, Side Satellite Sensor 3 (msec)	No
Clipping Time of Lateral Acceleration, Side Satellite Sensor 4 (msec)	SNA
Clipping Time of Lateral Acceleration, Floor Sensor (msec)	No



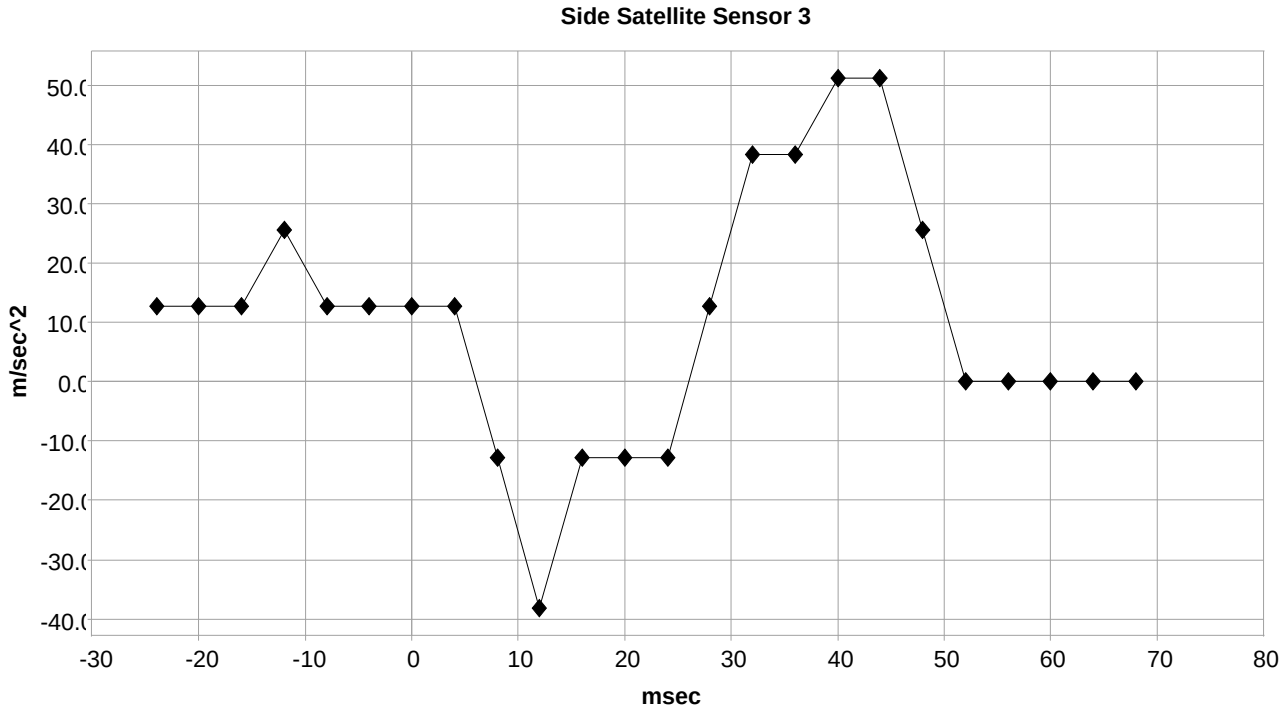
Deployment Time Marker Key

1	Driver/Passenger Pretensioner
2	Side Airbag
3	Rear Window Airbag Deployment Time



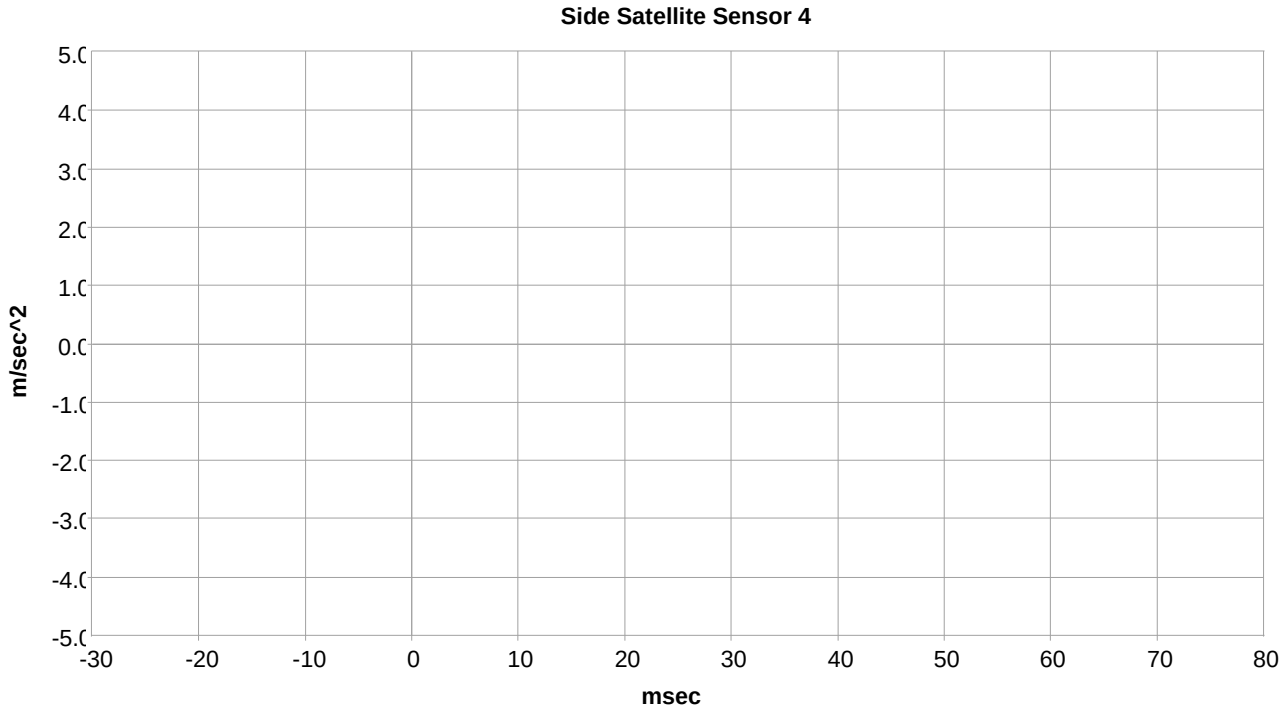
Deployment Time Marker Key

1	Driver/Passenger Pretensioner
2	Side Airbag
3	Rear Window Airbag Deployment Time



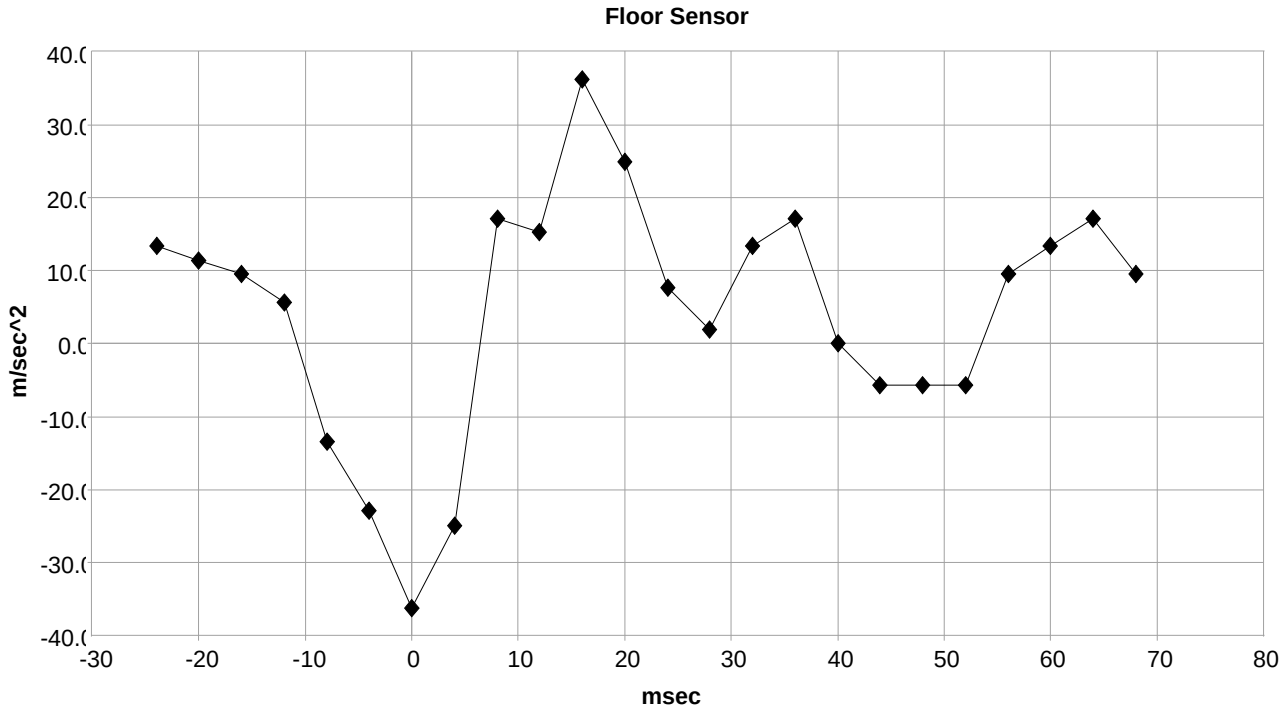
Deployment Time Marker Key

1	Side Curtain Airbag



Deployment Time Marker Key

1	Side Curtain Airbag



Deployment Time Marker Key

1	Driver/Passenger Pretensioner
2	Side Airbag
3	Rear Window Airbag Deployment Time
4	Side Curtain Airbag

Lateral Crash Pulse (4th Prior Event, TRG 2 - table 2 of 2)

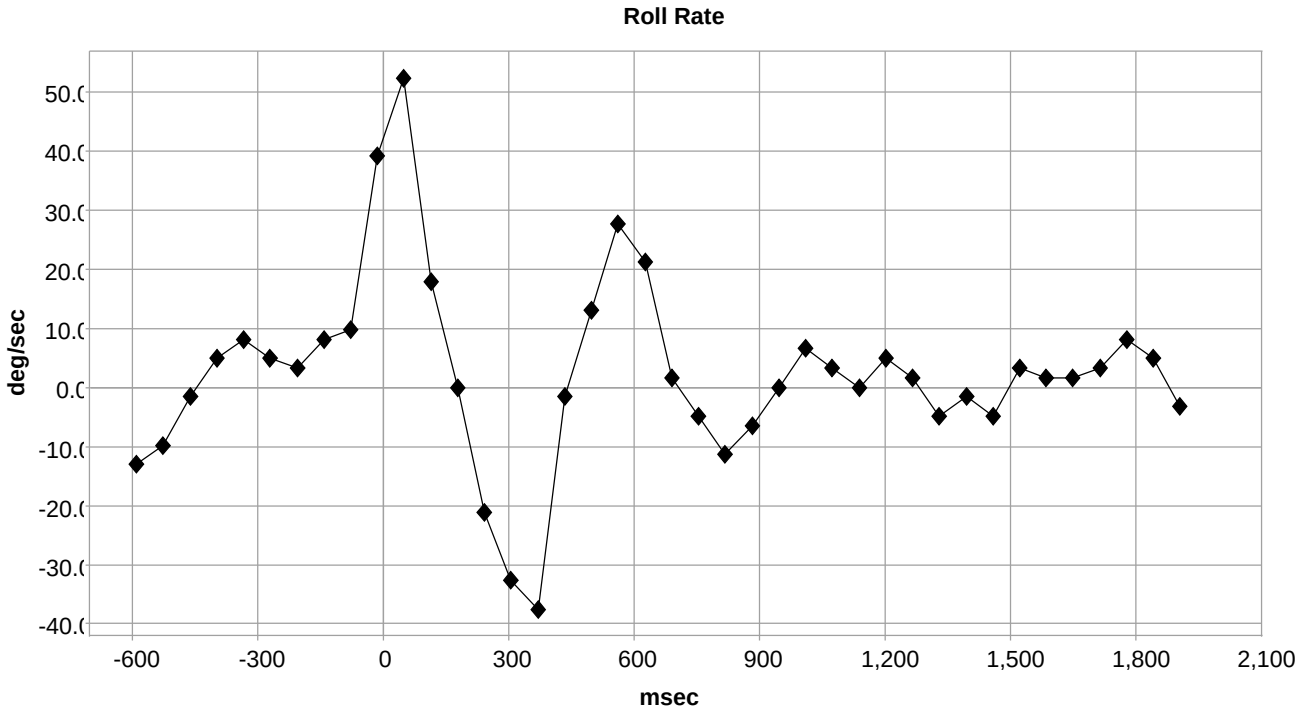
Time (msec)	Lateral Acceleration, Side Satellite Sensor 1 (m/sec^2)	Lateral Acceleration, Side Satellite Sensor 2 (m/sec^2)	Lateral Acceleration, Side Satellite Sensor 3 (m/sec^2)	Lateral Acceleration, Side Satellite Sensor 4 (m/sec^2)	Lateral Acceleration for Side Crash, Floor Sensor (m/sec^2)
-24	12.8	SNA	12.8	SNA	13.4
-20	12.8	SNA	12.8	SNA	11.5
-16	25.5	SNA	12.8	SNA	9.6
-12	25.5	SNA	25.5	SNA	5.7
-8	12.8	SNA	12.8	SNA	-13.4
-4	0.0	SNA	12.8	SNA	-23.0
0	-12.8	SNA	12.8	SNA	-36.4
4	-12.8	SNA	12.8	SNA	-24.9
8	-12.8	SNA	-12.8	SNA	17.2
12	-12.8	SNA	-38.3	SNA	15.3
16	0.0	SNA	-12.8	SNA	36.4
20	12.8	SNA	-12.8	SNA	24.9
24	0.0	SNA	-12.8	SNA	7.7
28	0.0	SNA	12.8	SNA	1.9
32	38.3	SNA	38.3	SNA	13.4
36	38.3	SNA	38.3	SNA	17.2
40	25.5	SNA	51.1	SNA	0.0
44	25.5	SNA	51.1	SNA	-5.7
48	12.8	SNA	25.5	SNA	-5.7
52	12.8	SNA	0.0	SNA	-5.7
56	12.8	SNA	0.0	SNA	9.6
60	12.8	SNA	0.0	SNA	13.4
64	12.8	SNA	0.0	SNA	17.2
68	12.8	SNA	0.0	SNA	9.6

System Status at Event (5th Prior Event, TRG 1)

Recording Status, Rollover Crash Info.	Complete
Crash Type	Rollover
TRG Count (times)	1
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Side Curtain Airbag Deployment, Time to Deploy (msec)	No
Pretensioner Deployment, Time to Fire, Driver (msec)	No
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	No

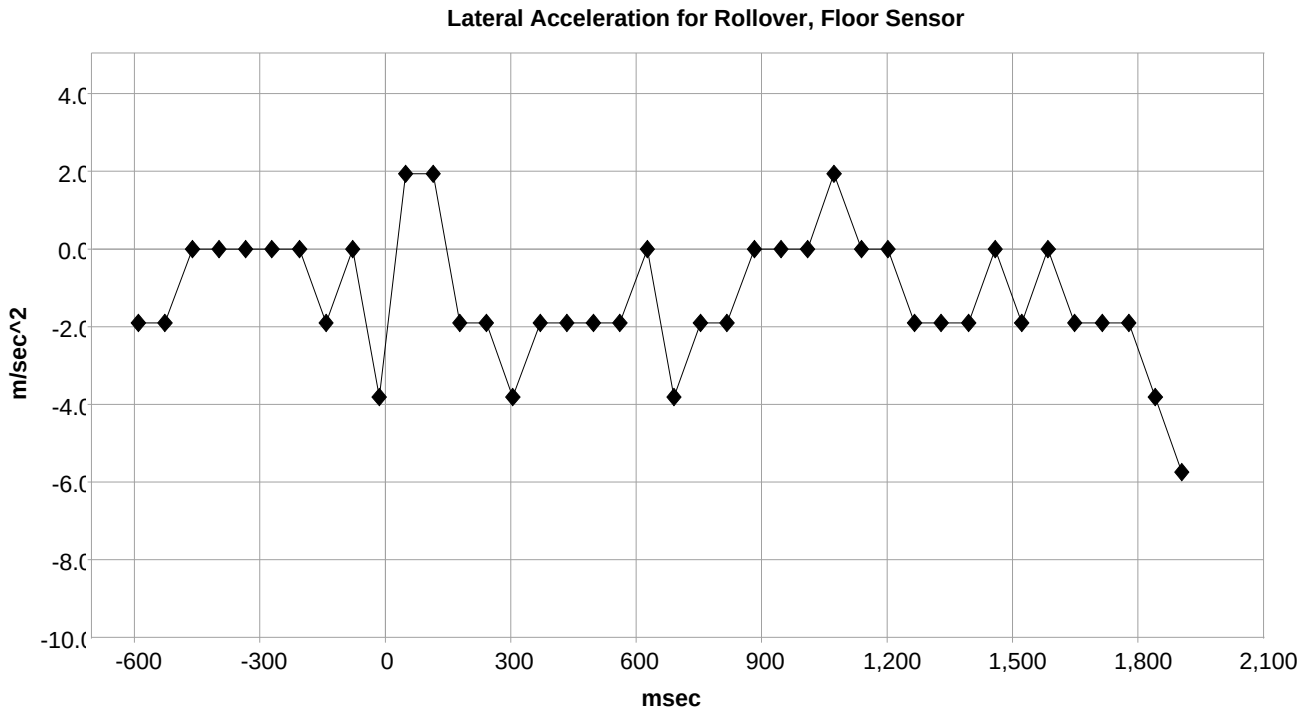
Rollover Crash Pulse (5th Prior Event, TRG 1 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	50
Roll Angle Peak (degrees)	5.2
Roll Angle at the Time of TRG (degrees)	2.6



Deployment Time Marker Key

1	Driver/Passenger CSA
2	Driver/Passenger Pretensioner



Deployment Time Marker Key

1	Driver/Passenger CSA
2	Driver/Passenger Pretensioner

Rollover Crash Pulse (5th Prior Event, TRG 1 - table 2 of 2)

Time (msec)	Roll Rate (deg/sec)	Lateral Acceleration for Rollover, Floor Sensor (m/sec^2)
-590	-13.0	-1.9
-526	-9.8	-1.9
-462	-1.6	0.0
-398	4.9	0.0
-334	8.1	0.0
-270	4.9	0.0
-206	3.3	0.0
-142	8.1	-1.9
-78	9.8	0.0
-14	39.1	-3.8
50	52.1	1.9
114	17.9	1.9
178	0.0	-1.9
242	-21.2	-1.9
306	-32.6	-3.8
370	-37.5	-1.9
434	-1.6	-1.9
498	13.0	-1.9
562	27.7	-1.9
626	21.2	0.0
690	1.6	-3.8
754	-4.9	-1.9
818	-11.4	-1.9
882	-6.5	0.0
946	0.0	0.0
1010	6.5	0.0
1074	3.3	1.9
1138	0.0	0.0
1202	4.9	0.0
1266	1.6	-1.9
1330	-4.9	-1.9
1394	-1.6	-1.9
1458	-4.9	0.0
1522	3.3	-1.9
1586	1.6	0.0
1650	1.6	-1.9
1714	3.3	-1.9
1778	8.1	-1.9
1842	4.9	-3.8
1906	-3.3	-5.7